

INTERNATIONAL SCHOOL

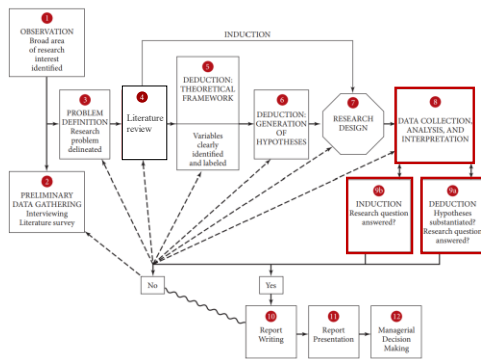
Quantitative & Qualitative Data Analysis

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RESEARCH WORKFLOW



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LEARNING OUTCOME

LEARNING OBJECTIVES:

1. Basic quantitative data analysis
2. Basic qualitative data analysis

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QUANTITATIVE DATA ANALYSIS

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INTRODUCTION

- After quantitative data have been collected from a representative sample of the population, the next step is to analyze them to answer our research questions.

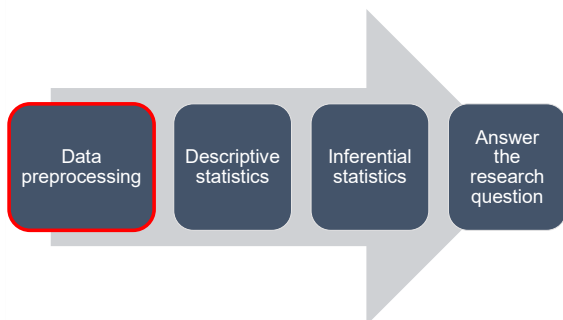


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QUANTITATIVE DATA ANALYSIS PROCESS

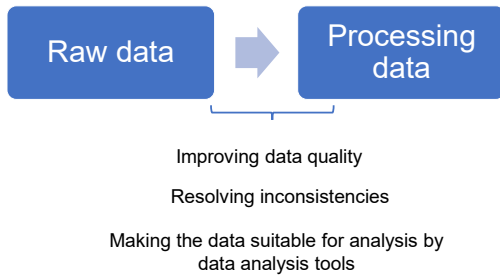


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Data preprocessing

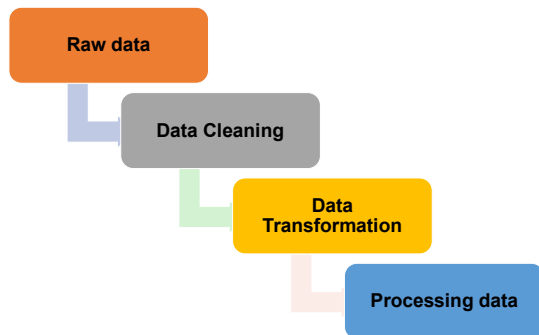


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Data preprocessing



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Data Cleansing

- Data cleansing involves **identifying and correcting errors** or inconsistencies in the data.



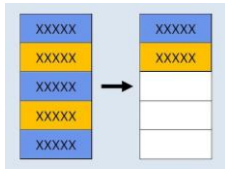
- It may include **removing duplicates**, **handling missing values** (e.g., imputation or deletion), and **dealing with outliers**.

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Data Cleansing



Removing duplicates

	col1	col2	col3
0	1.0	7.0	NaN
1	2.0	8.0	14.0
2	NaN	NaN	NaN
3	NaN	10.0	16.0
4	5.0	11.0	17.0
5	6.0	12.0	18.0

	col1	col2	col3
1	2.0	8.0	14.0
4	5.0	11.0	17.0
5	6.0	12.0	18.0

Removing missing values

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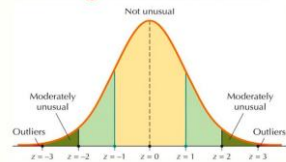
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Data Cleansing



Dealing with outliers

Detecting Outliers with z-Scores



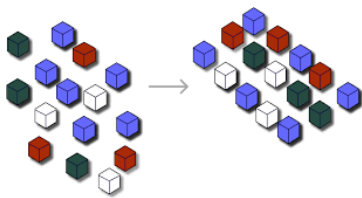
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Data Transformation

- Data transformation involves converting the data into a format that is suitable for analysis.
- Including Data coding and Encoding categorical variables.



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Data coding

- Data coding is a process in research where raw data is systematically organized, categorized, and labeled according to predefined criteria or coding schemes.

Age	Gender	Marital Sts	Highest education lev	Occupation	Is your occupation...
18	Female	Single	High school	Students	yes
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	University and above	Students	yes
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	yes
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	No
18	Female	Single	High school	Students	yes
18	Female	Single	University and above	Students	No

A1	A1group	A2	A4	A5	A6	A7	C
18	1	2	2	3	5	1	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	4	5	1	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	3	5	0	
18	1	2	2	3	5	1	
18	1	2	2	3	5	0	
18	1	2	2	3	5	1	
18	1	2	2	3	5	0	

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Encoding categorical variables.

- Encoding categorical variables is converting data from quantitative to qualitative variables (Categorical variables).

Age group	Age	Agegroup
	70	3
	28	1
	38	1
	39	1
	42	1
	40	1
	25	1
	26	1
	39	1
	30	1
	31	1
	41	1
	29	1
	64	2

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Data entry

- Data can be entered into a database through any software program
- Some software packages useful for data entry/analysis:
 - ✓ LISREL: from Scientific Software International
 - ✓ MATLAB®: from the MathWorks, Inc.
 - ✓ Mplus: developed by Linda and Bengt Muthén
 - ✓ Qualtrics: a private research software company
 - ✓ SAS/STAT: from SAS Institute
 - ✓ SPSS: from SPSS Inc.
 - ✓ SPSS AMOS: from SPSS Inc.
 - ✓ Stata: from Stata Corporation
 - ✓ MS Office (excel, word, PowerPoint): Microsoft

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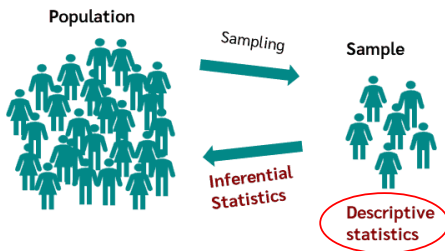
QUANTITATIVE DATA ANALYSIS PROCESS



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Descriptive statistics

Descriptive statistics are used to describe the characteristics of sample data.



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Descriptive statistics

Descriptive statistics for quantitative variables:

Measuring concentration

- Sum
- Number
- Mean
- Median
- Mode

Measure dispersion

- Data range
- Interquartile range (25%-75%) (Khoảng phân vị)
- Standard deviation (Độ lệch chuẩn)
- Variance (Phương sai)

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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- **codebook <variable name>**

-> Describe the number of a qualitative variable

		N
Age group	18-44	160
	45-65	43
	>65	9

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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- **proportion <variable name>**

-> Describe the percentage of a qualitative variable

		%
Marital status	Married	.3113208
	Single	.6556604
	Divorce	.004717
	Widow	.0283019

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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- **summarize <variable name>**

-> Describe the mean and standard deviation of a quantitative variable

	Mean	SD
Height (m)		

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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- `tab <variable name1> <variable name2>, col/row/cell`
-> Describe the number and percentage of two qualitative variables

		Male		Female	
		N	%	N	%
Highest education level	Primary school and below	1	1.30	1	0.74
	Secondary school				
	High school				
	University and above				

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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- `tabstat <variable name1>, statistics( mean sd ) by(<variable name2>)`

-> Describe the mean and standard deviation of a quantitative variable by a qualitative variable

	Male		Female	
	Mean	SD	Mean	SD
Weight (kg)	52.63636	6.664733	52.11852	6.356689

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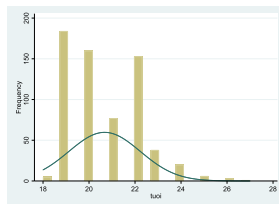
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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- `histogram <variable name>, freq normal`

-> Describe data range of a quantitative variable



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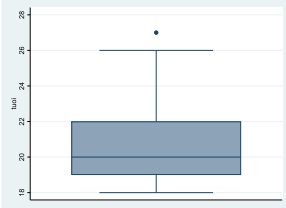
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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- **graph box <variable name>**

→ Describe data range of a quantitative variable



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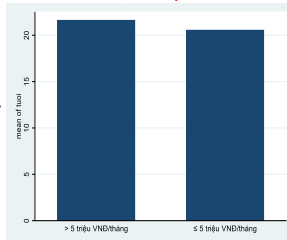
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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- **graph bar (mean) <variable name1>, over(<variable name2>)**

→ Describe the mean of a quantitative variable by a qualitative variable



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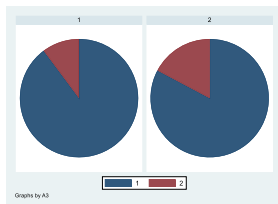
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Descriptive statistics

Some code to do descriptive statistics with **STATA**:

- **graph pie, over(<variable name>)**
- **graph pie, over(<variable name1>) by(<variable name2>)**



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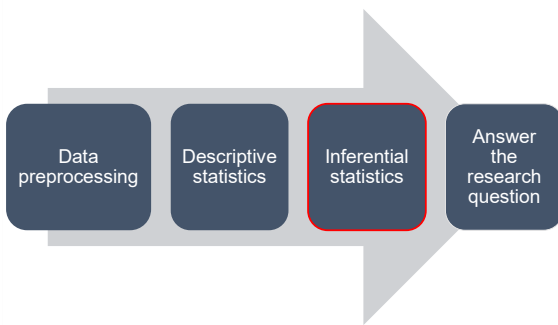
Practice analysis of descriptive statistics in
Tables 1, 2, 3 and 4

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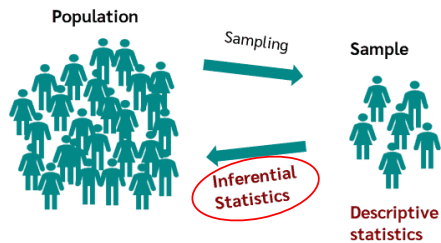
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Inferential statistics

Inferential statistics for quantitative variables:

- **Confidence interval** (Ước lượng khoảng tin cậy)
- **Hypothesis testing** (Kiểm định giả thuyết)



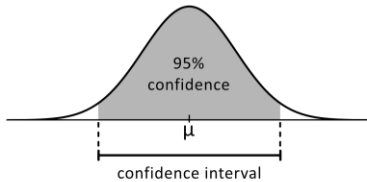
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Inferential statistics

- **Confidence interval** (Ước lượng khoảng tin cậy)
- Usually choose **95% confidence interval** (95%CI)
- 95% confident that the true **population value** is within the sample's confidence interval



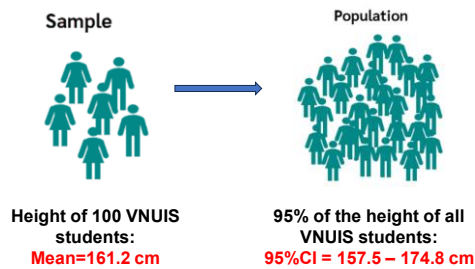
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Inferential statistics

- **Confidence interval** (Ước lượng khoảng tin cậy)



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Inferential statistics

- **Confidence interval** (Ước lượng khoảng tin cậy)

proportion <variable name>

→ Describe the percentage, SE and 95% CI of a nominal variable

ci means <variable name>

→ Describe the mean, SE and 95% CI of a continuous variable

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Inferential statistics

– Hypothesis testing

- Hypothesis testing is a statistical method that uses sample data to evaluate the validity of a hypothesis about a population parameter.
- Hypothesis testing, also called significance testing, aims to make a statistical conclusion about whether or not to accept the hypothesis.
- Usually, the statistical significance level is 95%, which means **p-value < 0.05** is considered a **statistically significant difference**.
- Example:** The height of VNUIS students has a **statistically significant difference** between **men** and **women**.

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Inferential statistics

– Hypothesis testing

Some basic statistical tests:

- Student t test
- ANOVA one-way test
- Chi-square test
- Pearson correlation

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Student t-test

- Compare the difference in **mean values** of **one or two independent groups**.
- Used for data with a continuous variable with normally distributed and a nominal variable has 1 or 2 subgroups.
- STATA code:** `ttest <continuous variable>, by (<nominal variable>)`

	Male		Female		P-value
	mean	sd	mean	sd	
Age					

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ANOVA one-way test

- Compare ≥ 3 average values of independent groups.
- Used for data with a continuous variable with normally distributed and a nominal variable has ≥ 3 subgroups
- **STATA code:** `oneway <continuous variable> <nominal variable>`

	Single		Married		Others		P-value
	mean	sd	mean	sd	mean	sd	
Mean of age							

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Chi-square test

- Compare the difference between **two or more ratios**.
- Used for data with two a nominal variable.
- **STATA code:** `tab <nominal variable1> <nominal variable2>, col chi2`

		Male		Female		P-value
		N	%	N	%	
Marital status	Married					
	Single					
	Divorce					
	Widow					

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Pearson correlation

- Compare the correlation of **two quantitative variables**.
- Used for data with two continuous variables.
- **STATA code:** `pwcorr <continuous variable1> <continuous variable2>`
- Values from **-1 to +1**
 - When $r > 0 \Rightarrow$ positive correlation
 - When $r < 0 \Rightarrow$ inverse correlation
 - The closer to 1 $\Rightarrow$ the stronger the correlation
- Convention:
 - $r < 0.3$: weak correlation
 - $r = 0.3-0.5$: average correlation
 - $r = 0.5-0.7$: strong correlation
 - $r = 0.7$: very strong correlation

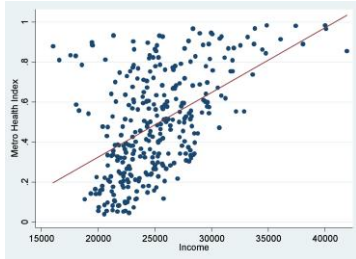
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Pearson correlation

$r = 0.89$ -> very strong correlation between income and metro health index



twoway scatter <variable name1> <variable name2> || lfit <variable name1> <variable name2>

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PRACTICE

Practice analysis of inferential statistics in
Tables 5, 6, 7 and 8

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